Retail Growth and Profitability Comparison

Amazon North America and Walmart U.S. | 2019-2024

Project purpose

This project compares the sales growth and operating profitability of an internet-origin retailer and a store-origin retailer. It uses 2019 as a pre-pandemic reference to describe how the two selected segments changed during 2020-2021 and the subsequent 2022-2024 period.

Data and approach

The sample contains twelve annual records: six for each segment. Segment net sales and operating income were manually transcribed from free official annual reports and embedded in one Jupyter notebook. Amounts are nominal USD millions. Amazon FY2019-FY2024 is paired with Walmart FY2020-FY2025. The fiscal periods are approximately aligned; actual reporting dates and source references are retained.

Four-module structure

Module	Main task	Class	Output
1. Research Configuration	Set the companies, years, reference year, comparison stages and study notes.	StudyConfig	Shared research settings
2. Data and Database Management	Clean amounts, align fiscal years, assign stages and save the records.	DatabaseManager	Financial records in SQLite and CSV
3. Metrics and Comparative Analysis	Calculate annual metrics, reference-year changes and stage means.	MetricCalculator	Annual results and stage summaries
4. Reporting and Workflow	Read saved results and create the comparison charts and report.	ReportBuilder	Three charts and an HTML report

Construction logic

The workflow is: official reports > embedded records > cleaned DataFrame > SQLite > calculated metrics > charts and report. Each module has one clear role. Small classes store the settings or output location they need, while ordinary functions handle cleaning and stage summaries.

Research content is separated from reusable processing code. Company details, fiscal-year adjustments and source records are entered in dedicated cells. To analyze different retailers, replace these settings and provide their real data; the same three-metric calculation and reporting framework can be reused.

Measures and core results

Metric definitions

Metric	Calculation
Annual sales growth (%)	(Current annual net sales / previous annual net sales - 1) x 100
Operating margin (%)	Segment operating income / segment net sales x 100
Sales index	Current annual net sales / reference-year net sales x 100

Each segment is compared with its own reference year. The 2019 sales index is 100, and 2019 sales growth is left missing because 2018 is outside the sample. Negative operating income is retained. Sales changes from the reference are cumulative percentages; margin changes are percentage points.

Comparison with the 2019 reference

Segment	Sales change, 2019-2024 (%)	2019 margin (%)	2024 margin (%)	Margin change (points)
Amazon North America	+126.91	4.12	6.44	+2.32
Walmart U.S.	+35.60	5.10	5.16	+0.07

Amazon North America expanded more strongly relative to 2019, while its operating margin followed a less stable path: it fell below zero in 2022 before recovering by 2024. Walmart U.S. grew more slowly, with its annual operating margin remaining close to 5%. Sales expansion and profitability therefore did not follow the same path.

Stage comparison

The study uses three groups: 2019 as the pre-pandemic baseline, 2020-2021 as the disruption period, and 2022-2024 as subsequent adjustment. Stage summaries use the arithmetic mean of available annual values and report the number of valid years for each metric. These means are not compound growth rates or pooled operating margins.

Outputs and interpretation

The notebook saves financial records, annual metrics and stage summaries in a dedicated SQLite database and three CSV files. It also produces sales-growth, operating-margin and sales-index charts, plus an HTML report. Company names and notes come from the configuration, while calculations use the saved financial values.

The findings are descriptive. There is only one pre-pandemic year, and the segments differ in geography, business mix and revenue recognition. The stage labels do not establish when the pandemic ended, and the observed differences cannot be attributed to the pandemic alone.

Official data sources

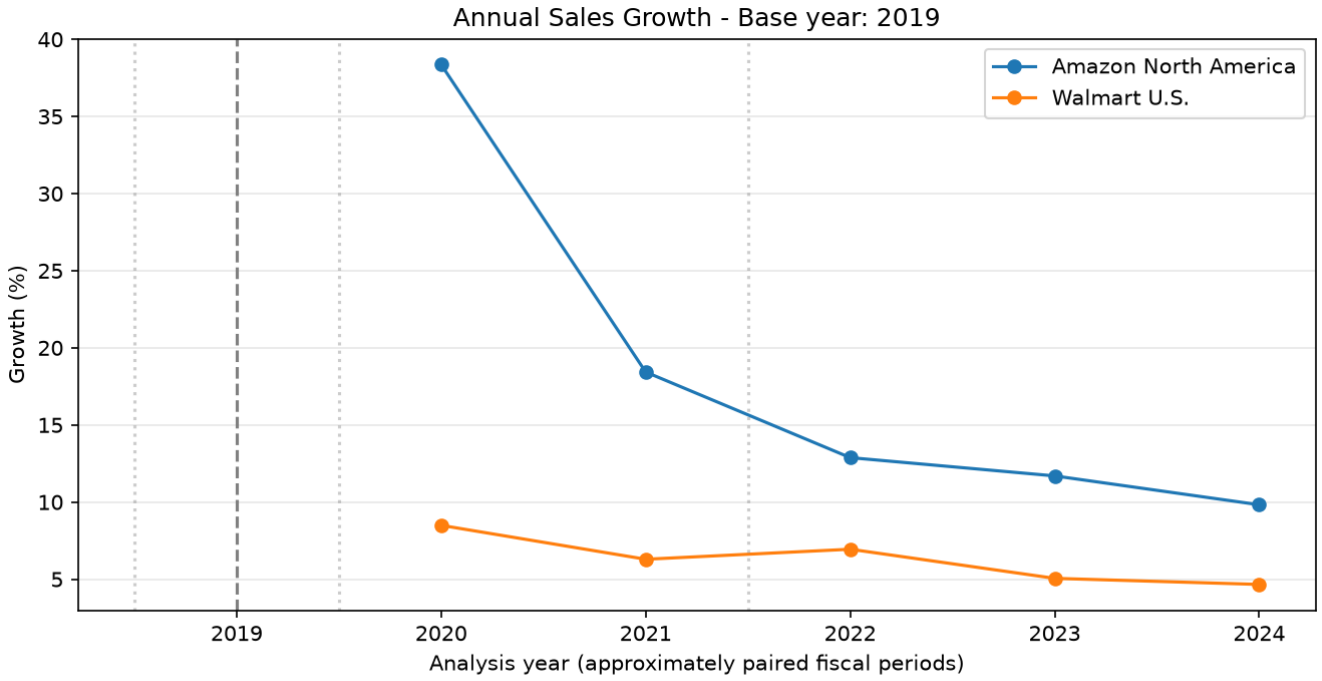
Amazon 2021 Form 10-K, Note 10: North America, 2019-2021.

Amazon 2024 Form 10-K, Note 10: North America, 2022-2024.

Walmart FY2022 Form 10-K, Item 7: Walmart U.S., FY2020-FY2022.

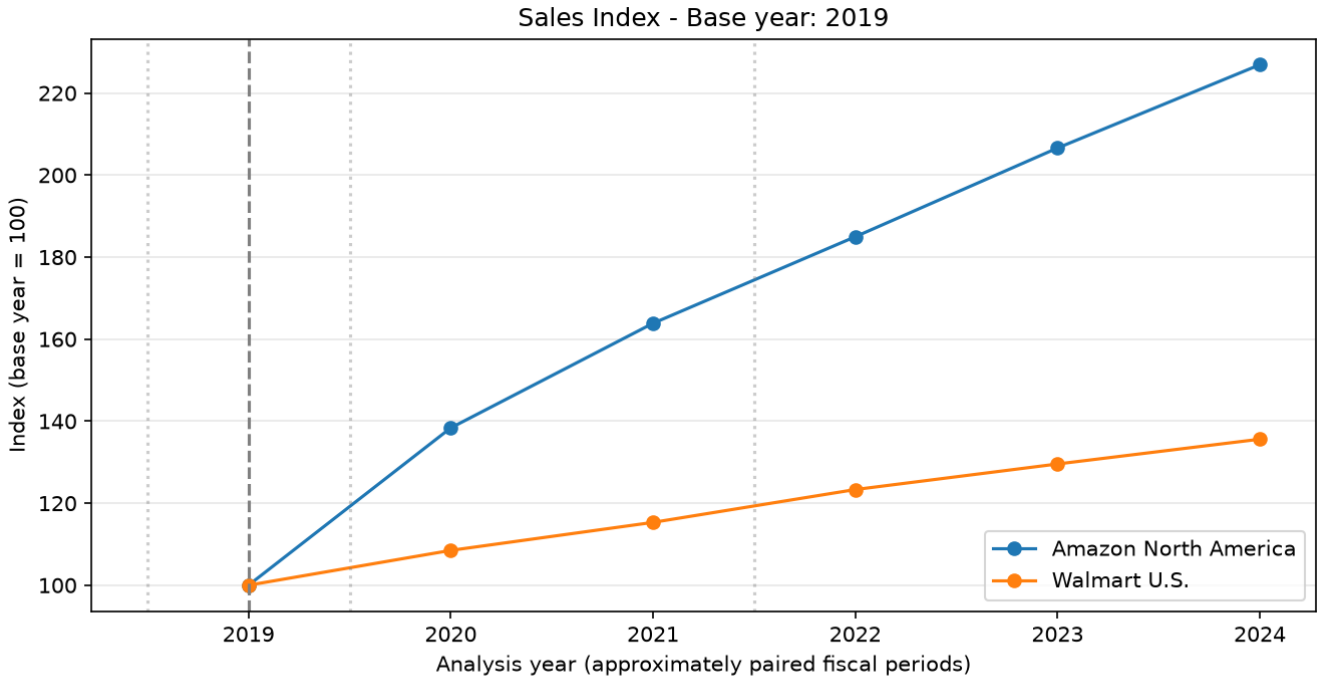
Walmart FY2025 Form 10-K, Item 7: Walmart U.S., FY2023-FY2025.

Sales growth and sales scale



Source: adopted official segment reports. Fiscal periods approximately paired.

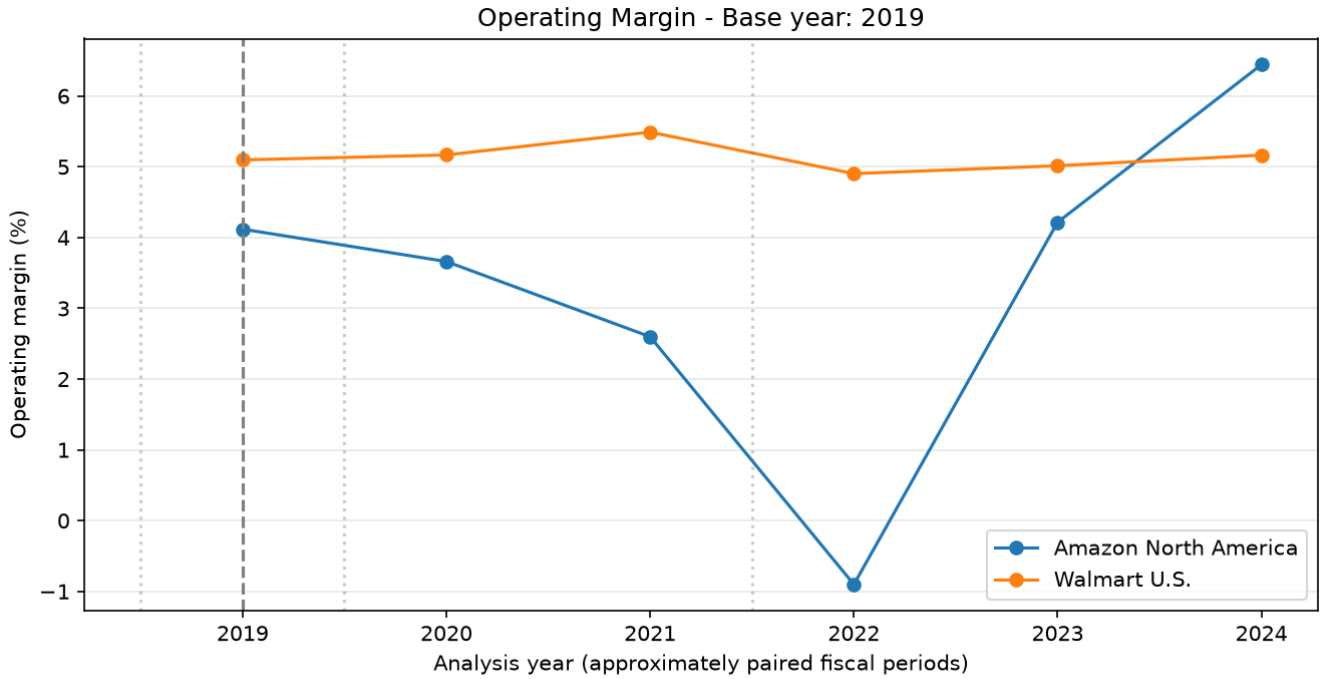
Figure 1. Annual sales growth. The 2019 growth value is unavailable, not zero.



Source: adopted official segment reports. Fiscal periods approximately paired.

Figure 2. Sales index, with each segment's 2019 net sales equal to 100.

Operating profitability



Source: adopted official segment reports. Fiscal periods approximately paired.

Figure 3. Operating income as a percentage of segment net sales. Negative values represent an operating loss.